Problem Set 2

Applied Stats/Quant Methods 1

Due: October 23, 2025

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday October 23, 2025. No late assignments will be accepted.

Question 1: Political Science

The following table was created using the data from a study run in a major Latin American city.[1] As part of the experimental treatment in the study, one employee of the research team was chosen to make illegal left turns across traffic to draw the attention of the police officers on shift. Two employee drivers were upper class, two were lower class drivers, and the identity of the driver was randomly assigned per encounter. The researchers were interested in whether officers were more or less likely to solicit a bribe from drivers depending on their class (officers use phrases like, "We can solve this the easy way" to draw a bribe). The table below shows the resulting data.

[1] Fried, Lagunes, and Venkataramani (2010). "Corruption and Inequality at the Crossroad: A Multimethod Study of Bribery and Discrimination in Latin America. *Latin American Research Review*. 45 (1): 76-97.

	Not Stopped	Bribe requested	Stopped/given warning
Upper class	14	6	7
Lower class	7	7	1

(a) Calculate the χ^2 test statistic by hand/manually (even better if you can do "by hand" in R).

```r
bribery_frequency_data <- matrix (c(14, 6, 7, 7, 7, 1), nrow = 2, dimnames = list(
  c("Upper Class", "Lower Class"),
  c("Not Stopped", "Bribe requested", "Stopped/Given Warning")
))
```

```r
observed_row_sum <- rowSums(bribery_frequency_data)
observed_column_sum <- colSums(bribery_frequency_data)
observed_total_sum <- sum(bribery_frequency_data)

bribery_frequency_expected <- matrix(c ((observed_row_sum[1] * observed_column_sum[1] /
                                          observed_total_sum),
                                        (observed_row_sum[2] * observed_column_sum[1] /
                                          observed_total_sum),
                                        (observed_row_sum[1] * observed_column_sum[2] /
                                          observed_total_sum),
                                        (observed_row_sum[2] * observed_column_sum[2] /
                                          observed_total_sum),
                                        (observed_row_sum[1] * observed_column_sum[3] /
                                          observed_total_sum),
                                        (observed_row_sum[2] * observed_column_sum[3] /
                                          observed_total_sum)
), nrow = 2, dimnames = list(
  c("Upper Class", "Lower Class"),
  c("Not Stopped", "Bribe requested", "Stopped/Given Warning")
))

#calculate the chi^2 test statistic manually
# chi^2 = sum of ( (frequency observed - frequency expected)^2 divided by frequency expected )

chi_sq_numerator <- (bribery_frequency_data - bribery_frequency_expected)^2
chi_sq_denominator <- bribery_frequency_expected
```

```
chi_sq_bribery_frequency_data <- sum(chi_sq_numerator / chi_sq_
    denominator)
```

The χ^2 test statistic is 3.4125

(b) Now calculate the p-value from the test statistic you just created (in `R`).[2] What do you conclude if $\alpha = 0.1$?

```
pchisq(chi_sq_bribery_frequency_data,
        df = 2,
        lower.tail = FALSE)
```

The p-value from the above test statistic is 0.1815453. With the provided $\alpha = 0.1$, we do not have evidence of dependence.

[2] Remember frequency should be > 5 for all cells, but let's calculate the p-value here anyway.

(c) Calculate the standardized residuals for each cell and put them in the table below.

```r
row_proportion <- (observed_row_sum / observed_total_sum)
column_proportion <- (observed_column_sum / observed_total_sum)

std_error_bribery_frequency <- matrix(
  c(
    sqrt(
      bribery_frequency_expected[1, 1] * (1 - row_proportion[1]) * (1 - column_proportion[1])
    ),
    sqrt(
      bribery_frequency_expected[2, 1] * (1 - row_proportion[2]) * (1 - column_proportion[1])
    ),
    sqrt(
      bribery_frequency_expected[1, 2] * (1 - row_proportion[1]) * (1 - column_proportion[2])
    ),
    sqrt(
      bribery_frequency_expected[2, 2] * (1 - row_proportion[2]) * (1 - column_proportion[2])
    ),
    sqrt(
      bribery_frequency_expected[1, 3] * (1 - row_proportion[1]) * (1 - column_proportion[3])
    ),
    sqrt(
      bribery_frequency_expected[2, 3] * (1 - row_proportion[2]) * (1 - column_proportion[3])
    )
  ),
  nrow = 2,
  dimnames = list(
    c("Upper Class", "Lower Class"),
    c("Not Stopped", "Bribe requested", "Stopped/Given Warning")
  )
)


standardized_residual_bribery_frequency <- (bribery_frequency_data - bribery_frequency_expected) /
  std_error_bribery_frequency
```

	Not Stopped	Bribe requested	Stopped/given warning
Upper class	0.4369314	-1.620185	1.389297
Lower class	-0.4369314	1.620185	-1.389297

(d) How might the standardized residuals help you interpret the results?

While the p-value does not provide sufficent evidence of dependence, the standardized residuals show no surprising deviations. Upper Class drivers are more likely to not be stopped, less likely to be asked for a bribe and more likely to be given a warning when stopped all suggest that the large p-value may be an artifact of sample size more than a lack of correlation.

Question 2: Economics

Chattopadhyay and Duflo were interested in whether women promote different policies than men.[3] Answering this question with observational data is pretty difficult due to potential confounding problems (e.g. the districts that choose female politicians are likely to systematically differ in other aspects too). Hence, they exploit a randomized policy experiment in India, where since the mid-1990s, $\frac{1}{3}$ of village council heads have been randomly reserved for women. A subset of the data from West Bengal can be found at the following link: `https://raw.githubusercontent.com/kosukeimai/qss/master/PREDICTION/women.csv`

Each observation in the data set represents a village and there are two villages associated with one GP (i.e. a level of government is called "GP"). Figure 1 below shows the names and descriptions of the variables in the dataset. The authors hypothesize that female politicians are more likely to support policies female voters want. Researchers found that more women complain about the quality of drinking water than men. You need to estimate the effect of the reservation policy on the number of new or repaired drinking water facilities in the villages.

[3] Chattopadhyay and Duflo. (2004). "Women as Policy Makers: Evidence from a Randomized Policy Experiment in India. *Econometrica.* 72 (5), 1409-1443.

Figure 1: Names and description of variables from Chattopadhyay and Duflo (2004).

Name	Description
`GP`	An identifier for the Gram Panchayat (GP)
`village`	identifier for each village
`reserved`	binary variable indicating whether the GP was reserved for women leaders or not
`female`	binary variable indicating whether the GP had a female leader or not
`irrigation`	variable measuring the number of new or repaired irrigation facilities in the village since the reserve policy started
`water`	variable measuring the number of new or repaired drinking-water facilities in the village since the reserve policy started

(a) State a null and alternative (two-tailed) hypothesis.

- H0 - the average number of water projects remains the same regardless of whether or not there were reserved seats for women leaders.
- Hα - the average number of water projects increases where there are reserved seats for women leaders.

(b) Run a bivariate regression to test this hypothesis in `R` (include your code!).

```
west_bengal_chatt_duflo_data <- read.csv(url("https://raw.githubusercontent.com/kosukeimai/qss/master/PREDICTION/women.csv") )
```

```

regression <- lm(water~reserved,  data=west_bengal_chatt_duflo_data)

# function for formatting the regression table
output_stargazer <- function(outputFile, ...) {
  output <- capture.output(stargazer(...))
```

```
  cat(paste(output, collapse = "\n"), "\n", file=outputFile, append=TRUE)
}

output_stargazer("./problemSets/PS02/my_answers/regression_output_reserved_water.tex", regression)
```

Table 1:

	Dependent variable:
	water
reserved	9.252**
	(3.948)
Constant	14.738***
	(2.286)
Observations	322
R^2	0.017
Adjusted R^2	0.014
Residual Std. Error	33.446 (df = 320)
F Statistic	5.493** (df = 1; 320)
Note:	*p<0.1; **p<0.05; ***p<0.01

(c) Interpret the coefficient estimate for reservation policy.

The β coefficient estimate of 9.25 tells us that on average, villages with reserved seats had substantially (9x) more water based projects undertaken.